C 21307 (Pages : 3) Name.....

Reg. No.....

FOURTH SEMESTER (CUCBCSS—UG) DEGREE EXAMINATION
APRIL 2022

Physics/Applied Physics

PHY 4C 04—ELECTRICITY MAGNETISM AND NUCLEAR PHYSICS

(2014—2018 Admissions)

Time : Three Hours Maximum : 80 Marks

Section A

Answer **all** *questions.*
Each question carries 1 *mark.*
Answer in a word **or** *phrase.*

1. Nuclear forces are appreciably bonly when the distance between nucleons is of the order of ————.
2. The particle emitted in β-decay together with electron is ————.
3. The ionization power is maximum for ————.
4. The principle used in cyclotron is ————.
5. An example for paramagnetic substance is ————.
6. Energy of cosmic rays is of the order of ————.
7. Value of electric charge and strangeness of d quark is ————.
8. The susceptibility of paramagnetic material is ————.
9. The colours of quarks are ————.
10. The principle is ———— used in the construction of atom bomb.

(10 × 1 = 10 marks)

Section B

Answer **all** *questions.*
Each question carries 2 *marks.*
Answer in two **or** *three sentences.*

11. Explain critical magnetic field Hc in superconductivity. How it is related to T_c ?
12. Give two examples for ferromagnetic materials.

Turn over

13. State Gauss Law.

14. List out the properties of electric lines of force.

15. Distinguish between primary and secondary cosmic rays.

(5 × 2 = 10 marks)

Section C

Answer any **four** *questions.*
Each question carries 5 *marks.*
Answer in one paragraph.

16. Distinguish between Para, Dia and ferromagnetic materials with examples.

17. Differentiate between nuclear fission and fusion with example.

18. Explain latitude and longitude effect in cosmic rays.

19. Explain the terms, decay constant, half life and average life as applied to a radioactive substance. Find the relation between them.

20. Give the construction, theory and limitations of cyclotron.

21. Write the fundamental laws of radioactivity.

(4 × 5 = 20 marks)

Section D

Answer any **two** *questions.*
Each question carries 5 *marks.*

22. A metallic wire 2 mm. in diameter carries a charge of 200 C in one hour.The metal contains 6×10^{22} free electrons per cubic centimetre .Calculate the current in the wire and the drift velocity of electrons.

23. A cyclotron in which the flux density is 1.4 weber/m^2 is employed to accelerate protons. How rapidly should the electric field between the Dees be reversed ? Mass of the proton = 1.67×10^{-27} kg. and charge 1.67×10^{-19}.

24. The radius of Ho^{165} is 7.731 fermi. Deduce the radius of He^4 .

25. A reactor is developing energy at the rate of 3000 kW. How many atoms of U^{235} undergo fission per second ? How many kilograms of U^{235} would be used in 1000 hours of operation assuming that on an average energy of 200 MeV is released per fission ?

26. Complete the reaction :

$\Pi^+ \longrightarrow$ .....

$\Pi^- \longrightarrow$ .....

$\Pi^0 \longrightarrow$ .....

$\mu^+ \longrightarrow$ .....

$\mu^- \longrightarrow$ .....

(2 × 5 = 10 marks)

Section E

Answer any **two** *questions.*

Each question carries 15 *marks.*

27. With the help of neat diagram and necessary theory,explain how the temperature co-efficient of the material of a resistor can be determined using Carey Foster's Bridge.

28. State and Prove Gauss Law. Find the electric field due to similarly and oppositely charged two infinite plane parallel sheets. Deduce the expression of a parallel plate capacitor, when the dielectric slab is inserted between the plates. Assume the slab thickness is less than the plate separation.

29. What are cosmic rays ? State the main features of these rays. Discuss the effect of earth's magnetic field, latitude and longitude on them. What is the possible origin of these rays ?

(2 × 15 = 30 marks)